A Global Meta-Analysis of Soil Salinity Prediction Integrating Satellite Remote Sensing, Soil Sampling, and Machine Learning

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***Abstract*— Despite the growing interest among researchers, satellite-based prediction of soil salinity remains highly uncertain. The improvements in prediction accuracy reported in previous studies are usually limited to a single area. We performed a meta-analysis of regional satellite-based soil salinity predictions combined with *in situ* soil sampling and machine learning. Based on R^2 and root-mean-square error (RMSE) collected, we evaluated the effects of various features on the model accuracy and established a Bayesian network to evaluate the joint causal effect of multifeatures. Most significant differences were found in soil sampling schemes and characteristics of the study area, including the mean and variability (averaged R^2 of 0.75 for soil sample sets with lower salinity variation and 0.62 for others) of the salinity, climate type (R^2 of 0.64 in arid areas and 0.74 in others), soil texture (R^2 of 0.66 in sandy areas and 0.57 in others), and the interval between sampling date and satellite data acquisition date (R^2 of 0.53 under the condition of over 15 days and 0.65 in others). Generally, using different satellite data has limited effects on model performance among which Sentinel-2 performed better ($R^2 = 0.72$) than Landsat ($R^2 = 0.66$). The sampling of subsamples for each sample should focus on their subpixel-scale spatial heterogeneity across satellite data rather than the number of subsamples. It is also necessary to select appropriate vegetation and salinity indices for different satellite data under different vegetation conditions. Among algorithms, random forests ($R^2 = 0.70$) and support vector machines ($R^2 = 0.71$) performed best.**

***Index Terms*— Hyperspectral, machine learning, multispectral, remote sensing, satellite, soil salinity.**

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NOMENCLATURE

NIR	Near-infrared band.
SWIR	Short-wave infrared band.
UAV	Unmanned aerial vehicle.
SSC	Soil salt content.
ECe	Electrical conductivity of saturated soil extract.
EC1:5	Electrical conductivity of a 1:5 soil–water dilution ratio.
SEM	Structural equation modeling.
BN	Bayesian network.
PRISMA	Preferred reporting items for systematic reviews and meta-analyses.
EM	Expectation–maximization.
MI	Mutual information.
FVC	Fractional vegetation cover.
XGBOOST	Extreme gradient boosting.
RF	Random forest.
MLR	Multiple linear regression.
ANN	Artificial neural network.
SVM	Support vector machine.
PLSR	Partial least squares regression.
SD	Standard deviation.
NDVI	Normalized difference vegetation index.
EVI	Enhanced vegetation index.
SAVI	Soil-adjusted vegetation index.
CRSI	Canopy response salinity index.
SI	Salinity index.
VI_SI	Vegetation and salinity index.

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I. Introduction

SOIL salinization is a global ecological problem leading to soil degradation; it is considered one of the main factors limiting the sustainable development of irrigated croplands [1], [2] and other economic activities [3]. Soil salinity is dominated by a combination of various natural and anthropogenic factors, such as landform [4], [5], land surface altitude [6], soil drainage condition [7], irrigation, climate, and soil texture. The rapid and accurate regional prediction of soil salinity is essential for decision-makers to formulate optimization policies for regions with soil salinization risks, which can avoid unreasonable use of water and land resources in irrigated areas as well as continuous degradation of land ecosystems. In arid areas, excessive irrigation will cause the shallow groundwater level to rise and cause secondary salinization if the drainage conditions are poor [8]; 20% of irrigated land over the world is affected by salinity, especially in Australia, China, Egypt, Uzbekistan, India, Iran, Iraq, Mexico, Pakistan, and so on [9]. Due to the increasing population and food demand, the potential increasing irrigation in arid/semiarid regions may result in a higher risk of salinization in more croplands in the future. Due to its advantages in providing regional and dynamic information based on imagery, satellite remote sensing, such as multispectral, hyperspectral, and active microwave images, has become an important approach to monitor regional soil salinity [2], [10]. One of the most commonly used approaches is establishing a regression model of pixel-scale image data and ground *in situ* salinity of soil samples. Subsequently, the established model is used to estimate soil salinity at a regional scale based on the reflectance value of the images. With the increasing popularity of machine learning approaches in the field of remote sensing application research in recent years, a variety of machine learning algorithms, such as RF, SVM, and ANN, have enabled us to incorporate more satellite imagery variables and auxiliary data such as terrain, vegetation, soil texture/moisture, and hydrological data into the regression model to fit the nonlinear relationship with the salinity of soil samples. Such studies using various methods and satellite remote sensing data have been applied to salt-affected areas around the world [11]–[16]. However, the comprehensive and systematic evaluation of how to improve their performance and universality remains limited. Researchers have made considerable efforts to improve the accuracy of soil salinity prediction using machine learning and remote sensing at the regional scale. Most published papers prove the effectiveness of their proposed improvements by comparing the accuracy of the developed predictive model with the existing models. However, the improvements reported in these studies are usually limited to a single area. In these comparisons, it is still difficult for us to promote a general guideline for selecting suitable remote sensing data, features, and models. In addition, in many cases, even when comparing similar modeling methods, different studies may report conflicting results, and it is challenging to obtain general recommendations with these isolated individual studies. Therefore, questions such as "Which remote sensing data is the best?" and "how can we effectively improve the prediction accuracy of soil salinity?" have not been answered. Therefore, we should synthesize the collective knowledge of peer-reviewed journal articles, to determine which remote sensing data, feature selection, machine learning models, and other factors can provide the most promising contributions to improve the prediction accuracy and under which conditions are they more effective.

The factors that lead to differences in the performance of such models are diverse, including the used satellite images, soil sampling schemes, characteristics of the study area, and algorithms that are described in the following.

1) *Satellite Images:* Multispectral satellite sensors were the preferred method for mapping and monitoring soil salinity, largely due to the low cost of such imagery. The most commonly used multispectral remote sensing data source for soil salinity estimation was Landsat (e.g., TM, ETM+, and OLI) [17], [18] with a 16-day revisit period due to open access and the high resolution of 30 m. By contrast, Sentinel-2 [19]–[21] has a higher resolution of 10–20 m and a smaller temporal granularity of five days. Compared with Landsat and Sentinel, MODIS has the advantage of smaller time granularity to capture salinity dynamics and the disadvantage of much lower spatial resolution. The NIR and SWIR of these sensors are considered effective to distinguish the spectrum of saline soil [2], [10]. Also, various VI_SIs calculated from multibands have been used as input variables for the prediction model, but it is still unclear when they are more effective. The limitations of these multispectral remote sensing methods can be attributed but not limited to the relatively low spectral resolution, especially when applied to areas with the spectral mixture and subpixel spatial heterogeneity. Hyperspectral satellite data can purportedly be used to solve spectral mixture problems caused by insufficient spectral resolution of the traditional multispectral approach. Some hyperspectral data, such as EO-1 Hyperion and HJ-1, have been used for soil salinity estimation [22], [23]. However, the availability of these hyperspectral datasets may limit their large-scale applications. Besides, active (radar) microwave imagery, such as Sentinel-1 backscatter data, has also been used to estimate soil salinity [24], [25], which currently has shown its advantages in soil moisture retrieval [26]. However, the characterization of saline soils using radar imagery and complex dielectric constants determined by radar backscattering inversion techniques requires some soil moisture, as the measurements depend principally on dielectric constant and permittivity determined by moisture conditions and chemical and biological compositions [27]. UAV remote sensing has also been used for soil salinity estimation [28], and it may have advantages due to its high resolution. However, its ability to estimate salinity at a regional scale remains very limited compared with satellite remote sensing approaches.
2) *Soil Sampling Schemes:* The following conditions hold.
    a) Differences in soil sampling schemes affect model performance. In study areas of different sizes,

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the spatial density and representativeness of soil samples will affect the predictive capability of the established model. In addition, at the subpixel scale, different sampling schemes were used. In some studies, the salt content of a soil sample represents the salinity of one pixel, while in other studies, the average value of multiple (approximately 3 to 5) adjacent samples represents the salinity of one pixel or one plot to reduce errors and enhance representativeness.

b) For the study areas with multiple land use and cover types, it is important to consider the representativeness of soil sampling in advance.

c) The temporal representation of soil sampling should also be considered. The dynamic changes of topsoil salinity may be rapidly affected by fluctuations in hydrological and meteorological variables [2] and human activities such as irrigation. To minimize the effect of the difference in surface salt between the date of soil sampling and the date of acquisition of satellite images except for studies using multiyear data [29], [30], we need to consider the abovementioned factors synergistically in the soil sampling scheme. However, except for studies in which the soil sampling and remote sensing data acquisition are scheduled on the same day, the analysis of the errors caused by the time difference of the soil sampling and remote sensing data acquisition is still limited.

d) In terms of different soil salinity extraction methods, such as SSC, ECe, and EC1:5, the salinity values of the target variable of models were measured in different units (e.g., in dS/m and g/kg) or dimensions. It may be one of the reasons why it is difficult to compare the model accuracy of multiple studies.

e) The depths of sampling in different studies also vary. Although most are described as the same "estimation of surface salt," the actual sampling depths may be at 0–5 cm, 0–15 cm, 0–20 cm, and so on, which differ from the depth that can be detected by remote sensing.

f) In some sampling schemes, the salinity value of each sampling point or plot is represented by the mean value of several surrounding subsamples. This method was believed to be able to reduce the error, which considers the representativeness of the sample and the matching with the pixel of the satellite image. However, whether this approach is reasonable and how to weigh the representativeness, the number of subsamples in each sample, and the size of the plot (as the cost of the sampling) is still unclear.

3) *Characteristics of the Study Area:* The contributions of climate, topography, soil texture, and so on to topsoil salt accumulation vary with study areas. The drier the climate, the higher the topsoil salinity may be due to the extremely high potential evaporation and insufficient precipitation leaching. Soil texture may affect the topsoil salinity through the transport of soil water and salt. In low-lying terrain, salt may accumulate due to lack of drainage. To some extent, the model performance may also vary when applied to study areas with different characteristics. There are differences between applications in arid/semiarid areas and other areas due to different climates, vegetation coverage, human activities, and topsoil salt dynamics. The possibility to accurately estimate soil salinity varies with soil moisture content, salt pureness, coatings, and spectral contrast with other surface features (e.g., braided stream beds, eroded terrain surfaces with truncated soils, and nonsaline silt-rich structural crusts) of the study areas [2]. The interference of vegetation in the study area may also cause spectral confusion, especially in the green and red bands [31]. Therefore, there are probably different mechanisms of the surface salinity retrieval between salinized bare land with low vegetation coverage and salt-affected cropland with higher vegetation coverage. In salinized bare land, the spectral interference from vegetation may come from salt-tolerant halophytes mainly. In cropland, it mainly comes from the crop and varies with stages of the growing season. Due to the potentially stronger dynamics of vegetation and soil moisture, the complexity of soil salinity retrieval in cropland may be higher. Although some vegetation indices (e.g., NDVI, EVI, and CRSI) have also been used to indirectly infer salinity by monitoring canopy temperature and crop growth [18], 32], [33] based on the assumption of the negative correlation between soil salinity and crop growth with an assumed steady-state approach of soil salinity and make use of multiyear data to filter other factors that may be more transient in time such as irrigation water management, pest, and diseases, it is still uncertain how the real salinity dynamics may cause the failure of the steady-state assumption. Also, the topography variability, hydrometeorological variability, and the intensity of human activities (e.g., irrigation) of the study area may dominate the effectiveness of introducing auxiliary data (e.g., topographic variables derived from DEM). If the terrain of the study area is very flat, adding terrain data derived from DEM to the model may not provide additional information related to salt transport and thus cannot substantially improve the performance of the model.

4) *Algorithms:* Different algorithms may have their advantages when applied to fit the relationship between the salinity of soil samples and the covariates. For example, neural networks may have advantages in nonlinear fitting [34] and RF may avoid overfitting due to the introduction of randomness [35]. However, when focusing on fitting the relationship between soil salinity and covariates, it is still unclear which algorithm is more universal and why it performed better.

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The main purpose of this study was to conduct a meta-analysis of published satellite remote sensing predictions of regional soil salinity combined with *in situ* soil sampling and machine learning. We define "meta-analysis" as a method to systematically integrate results from peer-reviewed studies and, through statistical comparison, assess the patterns of effects (e.g., the statistical metric of prediction model performance) by satellite data type and other features of interest. Furthermore, we analyze the complex relationships among the multiple variables collected focusing on model performance. Compared with SEM [36], which has been used in meta-analysis to analyze the relationship between such multiple factors [37]–[39], BN may have advantages in integrating expert knowledge and inference [40], [41]. Therefore, in this study, we will use BN to analyze the causal relationships within the process of integrating soil sampling, remote sensing data analysis, and machine learning. The remainder of this article is organized as follows. We first detail the methods used to conduct the meta-analysis, including selection and data extraction from published studies and the subsequent analysis. Next, we present results of the evaluation of the model performance divided by satellite data type, soil sampling schemes, characteristics of the study area, and algorithms. Third, we analyze the joint probability effects of multiple features on the model performance based on the BN as a tool to guide future practical studies with the tradeoff between accuracy and cost considered. Finally, we discuss the implications of the findings for remote sensing prediction of soil salinity. The results of this summary provide some guidance on the expected performance of soil salinity prediction in various conditions. Moreover, the proposed method of using BN for causal multifeature meta-analysis can be referenced in other meta-analysis studies (e.g., a meta-analysis of remote sensing estimation of soil moisture).

II. Methodology

A. Protocol for Selecting the Sample of Articles

Based on the Scopus database, we designed and applied a general query on title, abstract, and keywords to include relevant articles with the "OR" operator applied among expressions in Table I. We followed PRISMA [42] in the selection process (Fig. 1). The used criteria to include articles are as follows.

1) Articles were limited to regional soil salinity prediction using satellite data. Articles mainly focusing on UAV sensors or ground *in situ* hyperspectral sensors were not included.
2) Only articles with soil salinity sampling reported were included.
3) Only articles using multivariable regression were included. Articles aimed at soil salinity-level classification were not included.
4) Only articles with determination coefficient R-square (R^2) of the validation or test step as the measure of model performance reported were included. R^2 is the proportion of the variance in the dependent variable that is predictable from the independent variable(s), which provides a measure of how well the observed outcomes are replicated by the model, based on the proportion of total variation of outcomes explained by the model. It was selected over other accuracy measures because it is most frequently reported and thus would result in a larger sample size of articles.
5) Only articles published in English journals were included.
6) Two global-scale studies [43], [44] were excluded because the number of used soil samples data and spatial scale was considerably larger than other included articles applied at the watershed scale (focused in this meta-analysis).

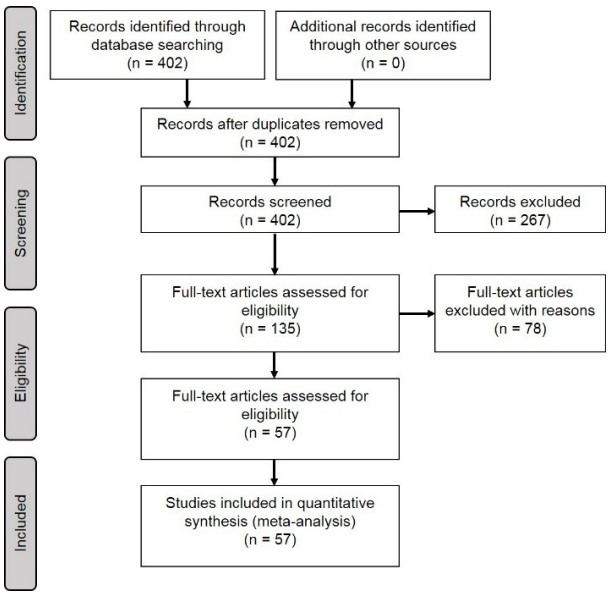


Fig. 1. PRISMA flow diagram for including articles.

B. Features of the Prediction Processes Evaluated

We consider a variety of features in Table I (see the Supplementary Material) involved in the soil salinity prediction process (Fig. 2) from the included papers. These variables can be roughly categorized into three groups: algorithms, remote sensing and auxiliary data, and characteristics of the study area and soil sampling. We recorded the algorithms used in the articles, and the performance of each model is processed as a data record. If multiple algorithms are applied comparatively to the same dataset and region, then multiple data records will be extracted. In the case of the same algorithm used, models that use different data or features will also be recorded as multiple records. The later meta-analysis is performed based on these records. R^2 (in the validation phase) is extracted as the main model performance evaluation indicator and the root-mean-square error (RMSE) is also included in the evaluation if reported in this article. However, the units and dimensions of RMSE for different salt measurement methods are different. Therefore, we use conversion factors to uniformly convert the RMSE, mean, maximum, and SD value of the salinity of soil sample collections to ECe with the unified unit dS/m. We extracted the average soil texture of soil samples from the SoilGrids1km [45] data of the basin-scale HydroSHEDS

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TABLE I
ARTICLE SEARCH QUERY DESIGN: '[A1 OR A2 OR A3…] AND [B1 OR B2…] AND [C1 OR C2…]'

ID	A	B	C
1	"soil salinity"	remote sensing	estimat*
2	"salinization"	satellite	predict*
3	"soil salt"	MODIS	retriev*
4	"salt-affected soil"	Landsat	model*
5	"soil electrical conductivity"	sentinel	regress*
6			"machine learning"
7			"Support Vector"
8			"Neural Network"
9			"Random Forest"

product in Table I (see the Supplementary Material) and the conversion factors are determined based on these extracted soil texture features in Table II (see the Supplementary Material) based on experience [46], [47]. As a result, most of the studies were given medium conversion factors because the sand content (%) and clay content (%) are not very high.

C. BN Used to Model the Joint Causal Effects of Multifeatures as a Tool Guiding Future Research

In addition to the above single analysis of the impact of each feature, it is necessary to simulate the causal relationship between model performance and the multifeature joint effects. We perform it using the BN for its potential to incorporate expert knowledge as a tool for guiding future research. Concretely, we process the multidimensional data collected from the articles into a training dataset with R^2 and RMSE as the target variables and other variables as the independent features. Based on empirical knowledge, we determine the directional causal link between nodes and the status discretization of each variable node. Furthermore, the BN is parameterized by the training data with the joint probability calculated.

The nodes of a BN represent random variables $(X_1,\ldots, X_n)$ with their joint probability distribution [48] calculated as

$$P(X) = P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \mathrm{pa}(X_i)) \quad (1)$$

where $\mathrm{pa}(X_i)$ represents the value of the parent node of the node X_i. With expert knowledge, the prior conditional probability table of BN can be given. Then, we use the EM algorithm [49] to incorporate observational data extracted from the included articles into the BN to update the conditional probability table.

To evaluate the sensitivity of model performance nodes to various characteristics, we use the sensitivity analysis of the BN with MI, which indicates the entropy reduction of the distribution of the child node caused by the determination of the status of the parent node. The greater the MI value, the higher the sensitivity of the child node to the parent node, accompanied by stronger causality. The formula of MI is as follows:

$$\begin{aligned} \mathrm{MI} &= H(Q) - H(Q|F) \\ &= \sum_{q}\sum_{f} P(q, f)\log_2\left(\frac{P(q, f)}{P(q)P(f)}\right) \end{aligned} \quad (2)$$

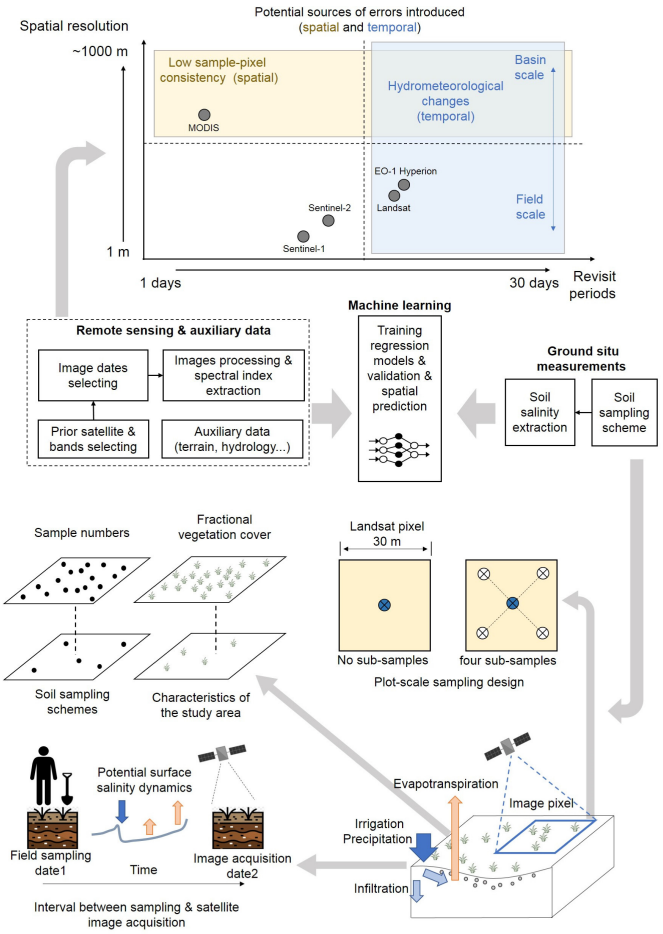


Fig. 2. Features of the soil salinity prediction process. The central part shows the general process of soil salinity prediction modeling using remote sensing, soil sampling, and machine learning methods. The upper part shows the potential sources of errors when using different remote sensing data. The lower part shows the factors that may cause variation in prediction accuracy at the stage of ground *in situ* soil sampling.

where H represents the entropy, Q represents the target node, F represents the set of other nodes, and q and f represent the status of Q and F, respectively.

We also develop another BN which included the studies using the Landsat data (most commonly used) only with a tradeoff analysis of cost and accuracy. The costs include transportation cost, soil analysis cost, and labor cost for sampling, and they are calculated as follows:

$$C_1 = aL + bn \quad (3)$$

$$C_2 = n\frac{Nd}{5 \times 30}p \quad (4)$$

$$C_3 = C_1 + C_2 \quad (5)$$

where C_1 is the sum of the transportation cost and soil analysis cost, C_2 is the labor cost for sampling, C_3 is the total costs, a is the gas bill per kilometer, b is the soil analysis cost per soil sample, L is the estimated total distance of the sampling event, n is the number of samples, N is the number of subsamples of each sample, d is the width of the sampling plot, and p is the labor cost under the five subsamples and

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TABLE II

TRADEOFF ANALYSIS BETWEEN ACCURACY AND COST UNDER DIFFERENT COMBINATIONS OF VARIABLE STATUS

FVC	study area	number of samples	VI_SI combinations	Change of the optimization goals in probability (%)		
				R^2 (0.8 to 1)	RMSE (0 to 5)	Cost_all (0 to 1000)
low	0 to 1000	0 to 80	SI	-3.4	-8.1	+32.8
low	0 to 1000	0 to 80	VI_SI	+1.6	-6.2	+32.8
low	0 to 1000	80 to 200	SI	-12.4	-5.6	-19.2
low	1000 to 10000	0 to 80	SI	+15.6	-14.2	+32.8
low	1000 to 10000	80 to 200	VI_SI	+20.6	-13.4	-13.2
low	1000 to 10000	80 to 200	SI	+18.6	-15.6	-13.2
medium	0 to 1000	0 to 80	SI	+0.6	+3.0	+32.8
medium	0 to 1000	0 to 80	VI_SI	-18.4	+18.2	+32.8
medium	0 to 1000	80 to 200	VI	-5.4	+12.3	-19.2
medium	1000 to 10000	0 to 80	VI	-0.4	-5.6	+32.8
medium	1000 to 10000	80 to 200	VI_SI	-4.4	+16.3	-13.2
medium	1000 to 10000	80 to 200	SI	-1.4	-1.2	-13.2
high	0 to 1000	0 to 80	VI	+12.6	-14.2	+32.8
high	0 to 1000	0 to 80	VI_SI	+3.6	-5.6	+32.8
high	0 to 1000	80 to 200	VI	+5.6	-6.2	-19.2
high	1000 to 10000	0 to 80	VI	-2.4	-10.7	+32.8
high	1000 to 10000	80 to 200	VI_SI	+10.6	-3.0	-13.2
high	1000 to 10000	80 to 200	VI	-5.4	-8.1	-13.2

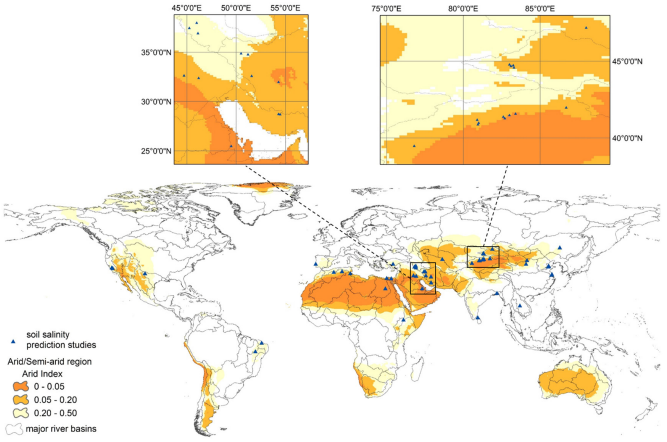


Fig. 3. Location of studies included in the meta-analysis ($n = 57$ experiments).

30×30 m plot condition. In fact, sampling and labor costs in different regions or research institutes of the world are different, so this study only uses this as an example of definite cost.

III. RESULTS

A. Articles Included in the Meta-Analysis

A total of 57 articles were selected and included in the meta-analysis during PRISMA flow (Fig. 1). The geographical coverage of these articles was mainly Asia (especially in Iran and Xinjiang, China), Africa, and North America (Fig. 3). Most of them are located in arid and semiarid regions with high potential evaporation and low precipitation. Irrigated agriculture in these areas is mostly affected by soil salinization (especially in the Middle East, Central Asia, and North Africa). Articles in nonarid areas are mostly near the sea, and salinization is caused by seawater intrusion into coastal groundwater. The number of published articles within this domain has increased rapidly in recent years (Fig. 4), and the combination of satellite remote sensing and machine learning for regional soil salinity prediction has attracted more researchers' interest.

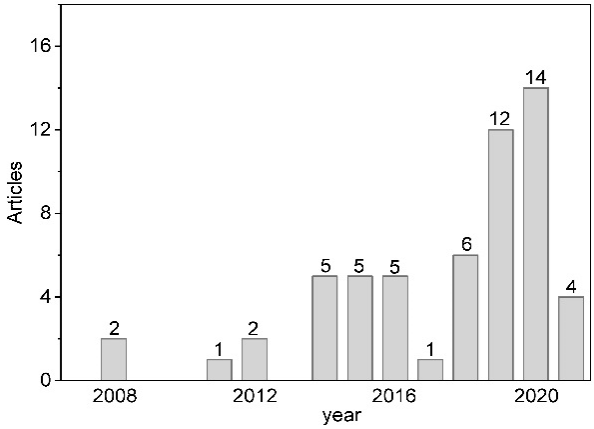


Fig. 4. Number of publications per year.

B. Meta-Analysis: Evaluation of the Effects of Various Features

Under the control of different types of characteristics, the difference in the distribution of R^2 is analyzed (see Fig. 5). These categories are divided into three groups focusing on algorithms, satellite and auxiliary data, and characteristics of the study area and soil sampling. The analysis results are as follows.

1) *In Terms of Algorithms:* The articles using Cubist showed the highest average R^2. Among algorithms that used frequently, RF and SVM performed better than ANN and MLR, while PLSR performed the worst. It also matches the results of internal comparisons of studies that developed multiple models based on the same datasets and model parameters (see Fig. 6) with the potentially disturbing effects of other features on the model performance are strictly limited.
2) *In Terms of Remote Sensing and Auxiliary Data:* The performance of models using different satellite data is different. Among the commonly used multispectral satellites, Sentinel-2 performed significantly better than Landsat which is most frequently used. The performance of the relatively rarely used MODIS was similar to that of Landsat. The models using active microwave radar data such as Sentinel-1 also performed well, and the performance was between Sentinel-2 and Landsat. The performance of the models using hyperspectral satellite data, such as HJ-1 and EO-1 Hyperion, is similar to that of Landsat. Unexpectedly, the models using multiple satellite data combinations did not show strong performance. Using a combination of SWIR and NIR is more effective than just using NIR. Besides, using hydrometeorological data and terrain data derived from DEM slightly improved the performance.
3) *In Terms of the Characteristics of the Study Area and Soil Sampling:* A medium S/n value corresponds to higher performance. The interval between the soil sampling date and the remote sensing data acquisition

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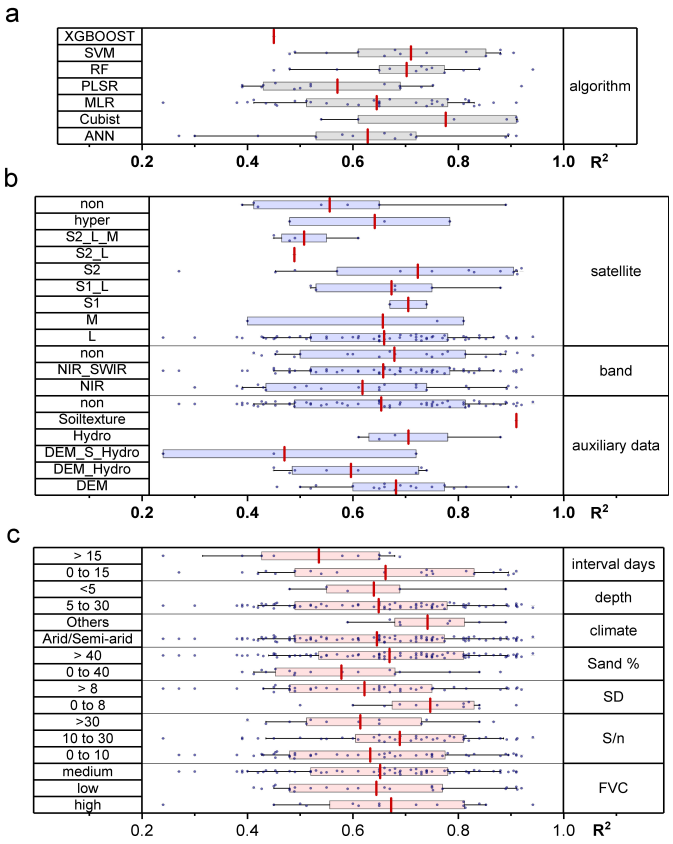


Fig. 5. Summary and test results of feature categories in the prediction process displayed in groups: (a) algorithms, (b) remote sensing and auxiliary data, and (c) characteristics of the study area and sampling. L, M, S1, S2, DEM, and Hydro respectively represent Landsat, MODIS, Sentinel-1, Sentinel-2, terrain data derived from DEM, and hydrometeorological data, respectively. Their combinations are indicated by an underscore (e.g., DEM_S_Hydro represents the combined usage of terrain data, soil texture, and hydrometeorological data). S/n is the area of the study area divided by the number of soil samples. FVC is the fractional vegetation coverage of the study area. In the box plots, the center lines represent the means, and the box limits represent the interquartile ranges.

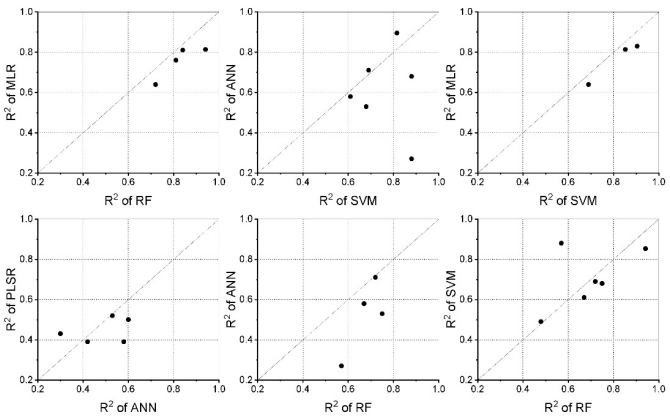


Fig. 6. Comparison of selected pairs of regression algorithms based on R^2. Regression algorithms: RF, MLR, ANN, SVM, and PLSR.

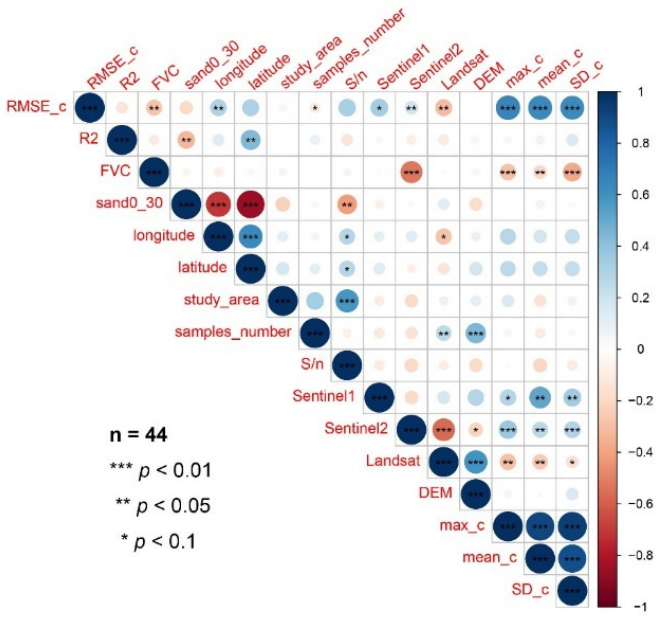


Fig. 7. Correlation matrix of evaluated features focusing on R^2 and RMSE. RMSE_c, max_c, mean_c, and SD_c are, respectively, the converted RMSE of models and the converted max value, mean value, and SD of salinity derived from soil samples, respectively. Sand0_30 is the sand content of the 0–30-cm topsoil. S/n is the value of the study area divided by sample numbers.

date significantly affected the model performance, and the subgroup with an interval greater than 15 days performed worst. It illustrates that the matching of the sampling date and the satellite image acquisition date is critical due to the correlation between the surface salt and the remotely sensed spectral data may decrease when the interval is more than 15 days. Therefore, when using satellite data with a revisit period of more than 15 days (e.g., Landsat), incorporating the revisit period and the projected quality of remote sensing images into the design of the sampling scheme may improve the model performance. When applied to collections of samples with higher mean salinity or its variation, the average model performed worse. For the sample collections with higher salinity, the correlation between salinity and spectrums may be lower. The relatively poor performance of models in arid and semiarid regions may be related to both high salinity and its variation, as well as stronger salt vertical transport due to the higher irrigation water volume/frequency and evaporation. The average sand content fraction and FVC in the sampling area are indirectly extracted from the literature with a certain uncertainty. Higher sand content corresponds to better model performance, which may be related to lower spectral interference of the topsoil moisture due to low soil water retention, and also, the salinity of topsoil with high sand content is inherently relatively low. Also, a higher FVC corresponds to a better model performance but not significantly. In addition, unexpectedly, no significant differences in model performance were observed under low, medium, and high FVC conditions.

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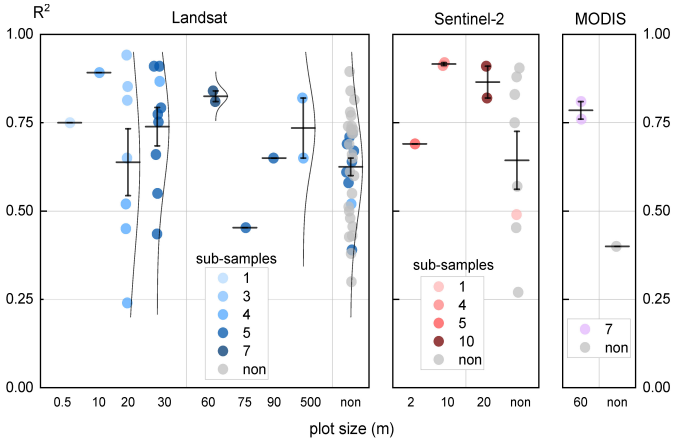


Fig. 8. Effect of the number of subsamples per soil sample and the size of the sampled plot (measured by its width) on the model performance using different satellite images. "non": information was not reported in this article.

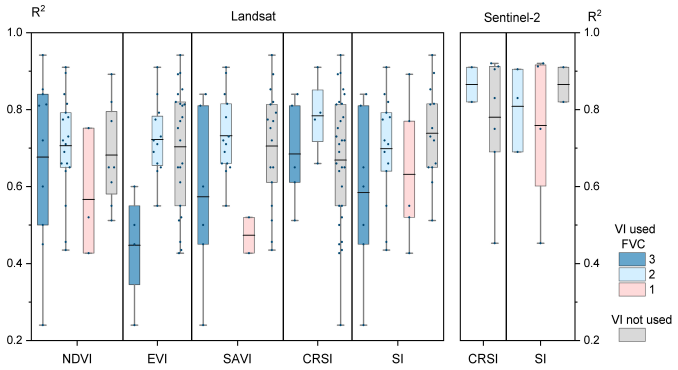


Fig. 9. Effect of the VI/salinity index calculated by the bands of different satellite images on the model performance when applied to the study area with different FVC. FVC is classified into one, two, and three levels, which represent low (mainly bare land), medium (a mixture of cropland and bare land), and high (mainly cropland) vegetation coverage, respectively. Note that records that did not use the five indices of NDVI, EVI, SAVI, CRSI, and SI are not included. In the box plots, the center lines represent the means, and the box limits represent the interquartile ranges.

On another level, we also evaluated the studies with R^2, the normalized RMSE, max, mean, and SD (of the salinity of the sample collection) at the ECe phase all recorded (Fig. 7). In a multiobjective evaluation criterion, features not only negatively correlated with RMSE but also positively correlated with R^2 are the most effective for improving the performance of the model. S/n satisfies this condition but not significantly (p-value $>$ 0.1). Also, the use of Landsat and FVC showed a significant negative correlation with RMSE, which means to improve the prediction of the model.

C. Meta-Analysis: The Potentially Applicable Plot Sampling Design and Vegetation/SI When Using Different Satellite Images

The selection of vegetation and SI in prediction model construction and a field-scale soil sample design is often more practical. On one hand, the number of subsamples of each sample and the area covered by the sampling plot may be related to the pixel size of the satellite image. Therefore, the effect of the sampling design of studies using different satellites was evaluated (see Fig. 8). When the sampling plot is larger, the number of subsamples is also larger. In the studies using only Landsat data, plots width smaller than 20 $\times$ 20 m^2 correspond to fewer than four subsamples, and plots larger than or equal to 30 $\times$ 30 m^2 correspond to more than five subsamples. In the studies using only Sentinel-2 data, plots width smaller than 10 $\times$ 10 m^2 correspond to fewer than five subsamples, and plots larger than or equal to 20 m correspond to more than ten subsamples. It shows that when the sample plot area is small (smaller than the size of satellite image pixel), researchers tended to use a small number of subsamples with lower costs. When the sampling plot reaches or exceeds one image pixel, researchers used more subsamples to try to reduce the error. When the sample plots were further increased (e.g., the 500 $\times$ 500 m^2 plots when using Landsat data), the number of samples did not increase accordingly. At the 30-m resolution of Landsat, the model performance of studies using five subsamples on the 30 $\times$ 30 m plot only improved a little than using 3–4 subsamples on the 20 $\times$ 20 m^2 plot. In summary, in terms of model performance, the effect of this improvement of multi subsamples may be insignificant. On the other hand, due to the difference in the wavelength range of different bands of satellite images, the evaluation of various vegetation/salinity indices was subdivided by different satellites (see Fig. 9). Among all VI_SIs, CRSI is the most useful for improving the model performance. Under the medium FVC conditions, the model performance is better than the low and high FVC conditions. Under medium and high FVC conditions, vegetation indices, such as NDVI, EVI, SAVI, and CRSI that simultaneously monitor vegetation conditions, performed better than SI, especially for studies using Landsat data. Under low FVC conditions, the performance of SI was better than other vegetation indices. Besides, CRSI and SI of sentinel-2 performed correspondingly better than CRSI and SI of Landsat under various FVC conditions. This shows that it is necessary to select appropriate vegetation and SI for different satellite data under different FVC conditions.

D. Joint Causal Effects of Multifeatures and the Tradeoff Between Accuracy and Cost Based on the BN

To build the BN with all records of included studies (BN_{all}), we constructed the causal relationship corresponding to the prediction process and quantified its conditional probability table using the data collected from the articles (see Fig. 10). R^2 and RMSE are regarded as the main target variables, and a variety of features are regarded as parent nodes that influence the target variable through the conditional probability formula. Based on experience and expert knowledge, we set the value range of each status of each feature node. Subsequently, the EM algorithm was used to compile the network. Based on the sensitivity analysis of BN_{all} (Fig. 12), we found that R^2 and RMSE are more sensitive to S/n, auxiliary data, and FVC, which is consistent with the results of the single feature analysis. With the joint influence of multiple features considered, the model performance is less sensitive to the satellite remote

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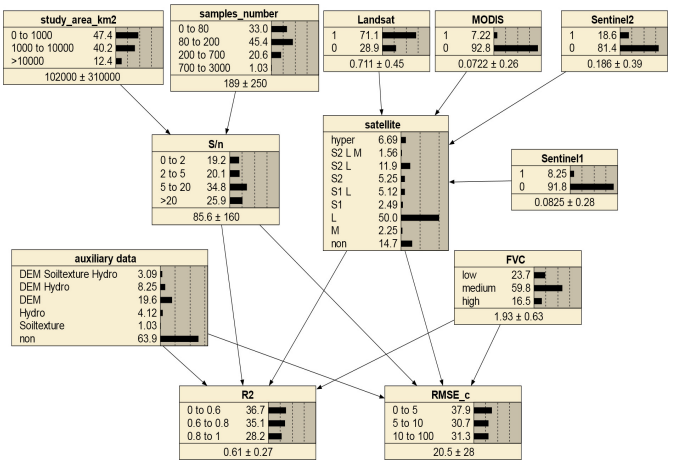


Fig. 10. Joint probability effects of features on the model performance (R^2 and RMSE) of the BN based on all studies (BN_{all}). The values before and after the "±" indicate the mean and SD of the distribution, respectively. S/n: study area divided by the number of samples. FVC: fractional vegetation coverage. L, M, S1, S2, DEM, and Hydro, respectively, represent Landsat, MODIS, Sentinel-1, Sentinel-2, terrain data derived from DEM, and hydrometeorological data.

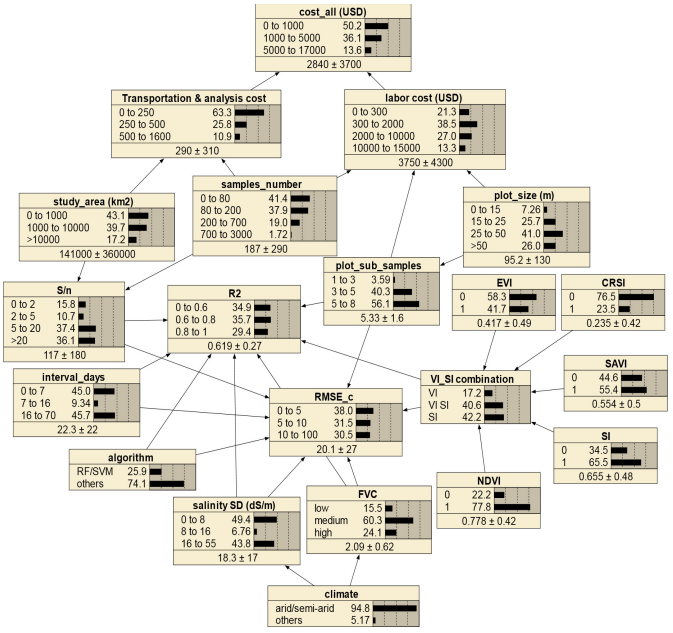


Fig. 11. Joint probability effects of features on the model performance (R^2 and RMSE) of the BN based on studies using Landsat data ($BN_{Landsat}$) with the tradeoff between accuracy and cost included. The values before and after "±" indicate the mean and SD of the distribution, respectively. S/n: study area divided by the number of samples. FVC: fractional vegetation coverage. VI_SI: vegetation and salinity indices.

sensing data used than other features. Among different satellite data, the model performance is more sensitive to Sentinel-2 and the least sensitive to MODIS.

Also, we further developed a more practical BN suitable for studies using Landsat data ($BN_{Landsat}$) (Fig. 11) with not only more detailed variables included (e.g., plot size, the number of subsamples, and the VI_SIs calculated with Landsat image bands) but also the tradeoff between accuracy and cost

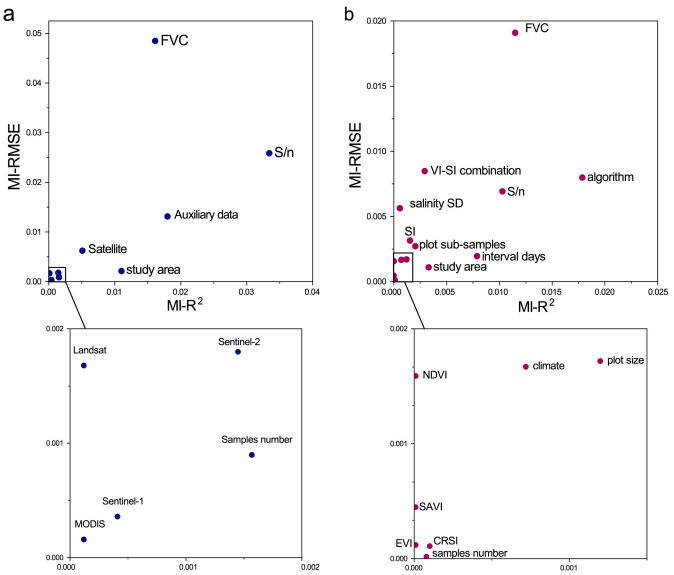


Fig. 12. Sensitivity of the R^2 and RMSE to the findings at various feature nodes of BN_{all} based on (a) all studies and (b) $BN_{Landsat}$ based on studies using Landsat data. MI: mutual information between nodes. MI-R^2 and MI-RMSE indicate the sensitivity of R^2 and RMSE to various nodes, respectively.

(transportation cost, soil sample salinity analysis cost, and field sampling labor cost). Similar to the BN_{all}, the R^2 and RMSE nodes of $BN_{Landsat}$ are more sensitive to FVC and S/n (see Fig. 12). Besides, they are also sensitive to the algorithm, the combinations of vegetation/salinity indices, the SD of the salinity of samples, and the interval days between the date of soil sampling and the satellite image acquisition. The R^2 node of $BN_{Landsat}$ is not significantly sensitive to the plot size and the number of subsamples.

In addition, the established BN can be used to guide future research as a tool to project the performance and cost of soil salinity prediction models. Using interactive operations and entering the findings of the nodes, we can infer that the posterior probability distribution of R^2, RMSE, and cost-related nodes with the status of some nodes determined (see Fig. 13). Changes in the probability distribution of nodes can help us recognize the potential impacts of various variables. We also performed a scenario analysis with $BN_{Landsat}$ to demonstrate the effectiveness of the BN in the analysis of the tradeoff between accuracy and cost under different combinations of variable status in Table II. It shows that the optimization of the target variables (e.g., accuracy and cost nodes) under different combinations of node status is not synchronized, possibly due to the potential tradeoff relationship.

IV. Discussion

Many articles have evaluated various satellite remote sensing data and machine learning algorithms to improve the prediction accuracy of regional soil salinity. To date, the results of these studies have not been combined and cannot provide a final guideline for the selection of the soil salt prediction process. Our work filled this gap by performing a meta-analysis of peer-reviewed studies, by statistically quantifying

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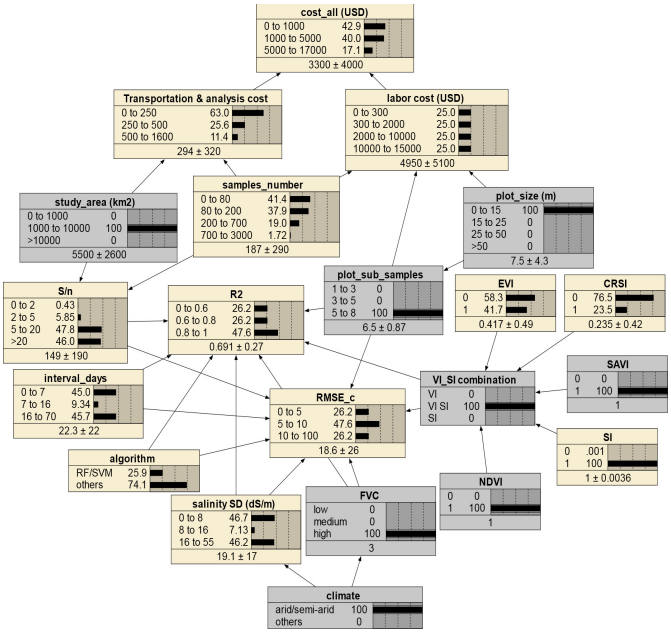


Fig. 13. Inference based on $tBN_{Landsat}$ with the status of some nodes determined.

the improvement in model performance achieved based on different features and regression algorithms. The current state-of-the-art technology was determined to provide pragmatic guidance from several years of published research, aimed at improving the accuracy of soil salinity prediction. We performed a meta-analysis of published regional satellite remote sensing predictions of soil salinity combined with *in situ* soil sampling and machine learning to better understand this approach and to guide future research. The effects of multiple features on the performance of the model were analyzed separately and jointly. Our analysis has shown that, despite the growing interest among researchers, satellite-based predictions of soil salinity remain uncertain, with potential for measurement errors. The main practical application of our results is to help researchers decide which improvement methods are most promising with the developed BN as an interactive and visual tool. With a clearer understanding of the expected improvements achieved through including different features, we can determine the priority of work and avoid practices that may decrease model accuracy in the actual modeling prediction process.

A. Opportunities and Challenges Exist for Improving Model Performance Based on the Main Findings of This Meta-Analysis

The applications of different satellite images have limited effects on the improvement of model performance, compared with the differences in the soil sampling schemes and the characteristics of the study area. It indicates that the effects of different temporal and spatial resolutions and spectral resolutions of various satellite remote sensing data on accuracy may be overestimated. As the two most commonly used multispectral satellite data for salinity prediction, Sentinel-2 performed a little better than Landsat probably due to higher spatial and temporal resolution [50]–[52]. The satisfactory performance of MODIS illustrates the importance of time granularity, which is also consistent with the significant impact of the interval between the sampling date and the remote sensing data acquisition date. When satellite data are used for soil salinity prediction, the small temporal granularity may be able to compensate for the negative effects of low spatial resolution. This also requires us to further study the spatial and temporal errors. The role of auxiliary data may also be limited especially in cultivated areas where topographical, hydrological, and meteorological variables may not be the dominant factors of soil salinity variation.

The most significant differences between model performances were found in different soil sampling schemes and characteristics of the study area. First, the higher the mean value and SD of the salinity of the sample collection, the poorer the model performance. This shows the difficulty to establish a quantitative regression relationship between high salinity values and covariates. Sample collections with high salinity variability may be accompanied by strong spatial heterogeneity, which may lead to strong spectral confusion and worse model performance [2]. Second, the temporal error cannot be ignored when the sampling date and the remote sensing data acquisition date are inconsistent. In previous studies, this kind of date interval occurred possibly due to the assumption of researchers that the surface salt is relatively stable within the interval, but this study shows that this may require more caution, due to surface salinity dynamics caused by human activities, hydrology, climate, and vegetation conditions especially in irrigated areas. Third, under different vegetation coverage conditions, the applicability of some VI_SIs is indeed different. For example, under low FVC conditions, the average performance of the SI, which pays more attention to the difference in the spectral reflectance of bare soil salinity, is better than other vegetation indices that pay more attention to vegetation conditions. With the rapid development of current machine learning methods, it may be still feasible to use some ineffective variables as input, but if the number of samples is limited, feature selection may still be necessary [53]. Fourth, the design of subsamples matching satellite image pixels may also have a certain meaning, but too many subsamples may be unnecessary, and more attention should be paid to the subpixel-scale spatial heterogeneity of the subsamples. It is also related to the cost because the cost of a large number of subsamples in each plot (especially at a large plot, such as the 500-m-wide pixel of MODIS) is significantly higher including labor costs and the cost of salt chemical analysis.

In summary, most of the current satellite data used for regional soil salinity prediction have certain practicability, and we may need to pay more attention to the matching of the soil sampling schemes and the characteristics of the study area

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with these satellite data with an in-depth understanding of the specific response mechanism of remote sensing data to the salinity variation, especially in terms of temporal and spatial granularity.

B. Choice of the Features Involves Tradeoffs Between Accuracy and Costs

In the practical implementation and prediction, we need to consider the sampling scheme, the remote sensing data, and the tradeoff of the expected accuracy and cost (e.g., satellite data processing cost, plot subsamples sampling cost, transportation-related cost, and soil sample analysis cost). The built BN that is based on previous studies can provide quantitative guidance with the practicality in modeling the joint effects of multifeatures on the model performance. Under the conditions of the combined multivariate changes, it can analyze the main influencing factors and effectively estimate the probability distribution of model performance with some status of feature nodes determined. Also, the BN can guide soil sampling schemes through diagnostic analysis. For example, if we only need a moderate accuracy, the BN can give recommendations for soil sampling density with different satellite data. The tradeoff between accuracy and cost may also vary with the research aims. Most articles included in this meta-analysis mainly focused more on achieving high prediction accuracy with high spatial resolution datasets. However, in practical agricultural management, our needs for the high spatial resolution may not be very essential because coarse-scale salinity prediction is enough to support agricultural management (e.g., determining the sowing area before the growing season). Although MODIS has disadvantages in spatial resolution, it has significant advantages over Sentinel-2 and Landsat in temporal resolution, which may be more effective in predicting the salinity dynamics with its time-series data [33] in the growing season.

C. Uncertainties and Limitations of This Meta-Analysis

The potential uncertainties and limitations of the results of this meta-analysis are as follows.

1) *Publication Bias and Weight:* This study did not pay much attention to publication bias because the number of articles that can be included in the meta-analysis is relatively limited. Previous meta-analysis studies were often performed based on the quality of the published journal and the public availability of research data [54], [55]. Unfortunately, most of the articles included in this study did not share soil sample data and established models publicly. The preferences in specific soil sampling and the setting of machine learning model parameters are not transparent, which makes it difficult to evaluate the reliability of the results of a single study, although most of the remote sensing data used were obtained from open source. Besides, the previous meta-analysis studies often weigh the effects of included studies based on sample size and variance of experimental results [56]–[58]. However, in this study, the number of samples was regarded as one of the evaluated features due to the lack of convincing methods to determine the weights between the models of the included articles.
2) *Limitations of the Criteria for Including Articles:* For the quantitative evaluation of model accuracy, we only selected studies that established multivariate regression models and reported R^2. Studies focusing on the classification of soil salinity levels were not included, and studies that established a univariate regression model were not included. It may lead to the incompleteness and limitation of the evaluation of some features. The remote sensing datasets used in the included articles are still mainly multispectral data and, thus, the evaluation of the studies using active/passive microwave remote sensing [59], [60], and hyperspectral data are insufficient. In addition, when considering the relationship between the measured soil salinity and remote sensing data, this study did not distinguish those studies that have used remote sensing data for multiyear [29], [30] focusing on the root zone but still calculated the time interval based on the remote sensing data acquisition date in the same year as the soil sampling. However, since there are only a few studies using data for multiyear, its impact on the evaluation results may be limited. Also, only articles published in English journals were included. However, many studies may have been published in other languages (e.g., Chinese, Spanish, and Arabic) because most areas affected by soil salinization are not English-speaking countries.
3) *Uncertainty of the Extracted Feature Information:* First, because most studies use far more soil sample inputs in the regression model than the covariates, we did not adapt R^2 in this study to the adjusted- R^2. Although this may cause partial overestimation of the positive effects of several features, second, the soil texture extracted from HydroSHEDS may also introduce errors when converted from the watershed to the field scale at the *in situ* soil sample sites. Therefore, based on the extracted soil texture, the conversion of the mean, maximum, SD, and model RMSE of the salinity measured with different methods may also introduce some errors. Thus, due to the uncertainty in the normalization of salinity values, the RMSE-based evaluation in this article may be biased compared to R^2. Third, the classification of FVC, in which the land use and land cover of the study area and the date of the soil sampling are comprehensively considered, may introduce errors due to subjectivity. Fourth, the interference effect of soil moisture [61] on the spectrum of salinized lands has not been specifically studied in this study because the *in situ* measured soil moisture data are almost not used in the selected published articles used for the meta-analysis. The temporal variability of soil moisture may be distinguished by the difference in satellite time resolution. For example, the five-day resolution of Sentinel-2 can contain less likely soil moisture fluctuations caused by precipitation

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or irrigation than the 16-day resolution of Landsat. The influence of different irrigation methods can also be partly attributed to the temporal soil moisture variation. Compared with flood irrigated areas, in drip-irrigated areas, soil moisture fluctuation may be smaller due to higher irrigation frequency and lower irrigation volume per time. Therefore, the possibility of model failure due to large soil moisture fluctuations can be lower than that in flood irrigated areas. Fifth, considering that a large number of studies are located in arid and extremely arid regions, low precipitation has little effect on the leaching of surface salt, so we did not distinguish between wet and dry seasons in this study. Sixth, since most of the soil salinity samples of the articles included are located in low-lying flat lands that are often more salt-affected, we did not consider the slope differences among these study areas. However, a few included studies may still include some sampling sites located in undulating terrains, which may lead to the overestimation of the effectiveness of topographic variables extracted from DEM data in the interpretation of soil salinity variation. Correspondingly, in a small area of cropland, the contribution of topographical variables to the improvement of model accuracy may be limited. For small-scale undulating terrain, the current commonly used DEM data with a resolution of 90 m may not be sufficient to capture the spatial heterogeneity of the subpixel salinity due to the coarse spatial resolution. Finally, various parameter and structure settings and optimization forms of the models are not treated differently. Instead, they were classified into algorithm families. It may also bring uncertainty to the subsequent evaluation.

4) *Independence Between Features:* In terms of the characteristics of the study area, although multiple features were extracted, the independence between these features was not taken into consideration. For example, mean salinity values of the sample collection, FVC, soil texture, and whether the study area belong to the arid/semiarid region that may not be independent, which may interfere with the evaluation of the effect of a single feature.
5) *Potential Expert Knowledge Integrated Into the BN:* In this study, the BN was completely parameterized based on the data collected from published articles. In future applications, expert knowledge in the field of remote sensing estimation of soil salinity can be first incorporated by giving the prior distribution (the relationship between multifeatures and the potential effects on accuracy) and then use the observation records of the included articles to update the BN to achieve the best integration and utilization of qualitative and quantitative information. With the inclusion of new records, the assessment objectives may change, and the structure of the BN may also need to be adapted.

V. Conclusion

We performed a meta-analysis of regional satellite-based soil salinity predictions combined with *in situ* soil sampling and machine learning. We evaluated the various features involved in the prediction process and established a BN based on multiple features to evaluate the joint causal effect and guide future research with the tradeoff between accuracy and cost. The main conclusions of this study are as follows.

1) Most significant differences between the model performance were found in different soil sampling schemes and the characteristics of the study area, including the mean and variability (averaged R^2 of 0.75 in sample sets with lower salinity variation and 0.62 in others) of the salinity of the soil samples, the climate type (averaged R^2 of 0.64 in arid/semiarid areas and 0.74 in others), the soil texture of the study area (averaged R^2 of 0.66 in sandy areas with sand content higher than 40% and 0.57 in others), and the interval between sampling date and satellite data acquisition date (averaged R^2 of 0.53 under the condition of over 15 days and 0.65 in others).
2) Using different satellite data has limited effects on the improvement of model performance among which Sentinel-2 (averaged $R^2 = 0.72$) performed better than Landsat (averaged $R^2 = 0.66$). The sampling of subsamples for each sample should focus on their subpixel-scale spatial heterogeneity across various satellite data rather than the number of subsamples. It is also necessary to select appropriate VI_SIs for different satellite data under different FVC conditions.
3) RF (averaged $R^2 = 0.70$) and SVM (averaged $R^2 = 0.71$) performed best among all evaluated algorithms.
4) The established BN has practicality in guiding future research with the tradeoff between accuracy and cost.

Acknowledgment

The built BN is publicly available at https://doi.org/10.5281/zenodo.5146240.

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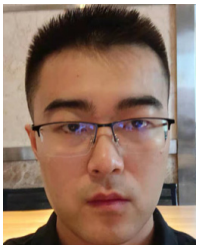
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